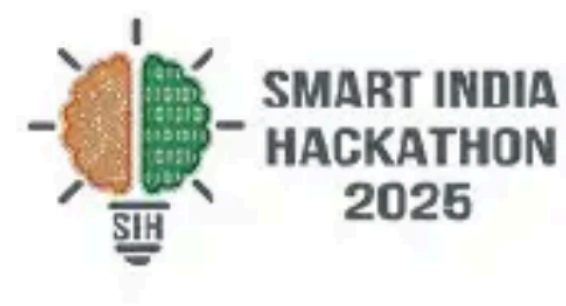
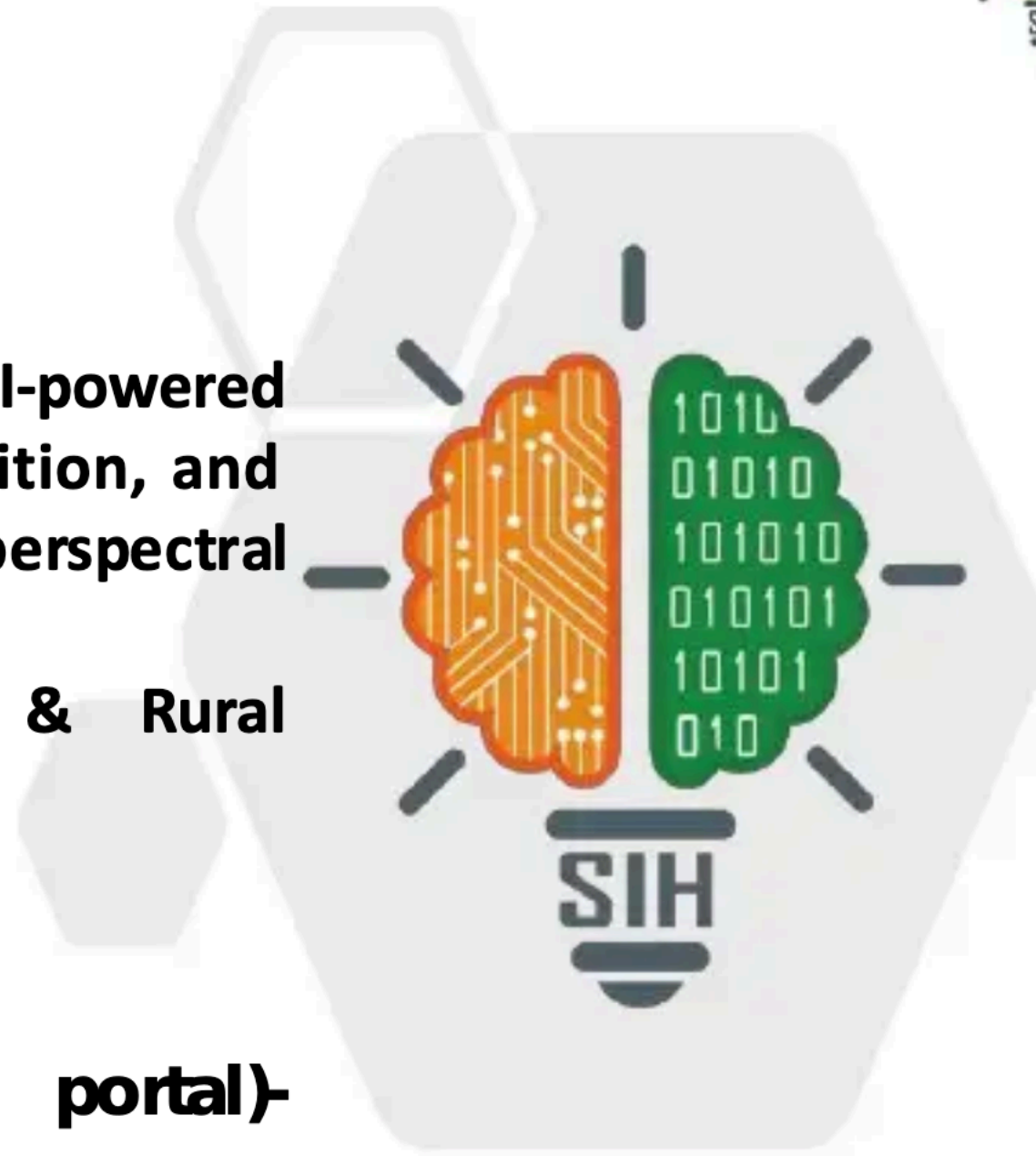


SMART INDIA HACKATHON 2025



- **Problem Statement ID- 25099**
- **Problem Statement Title-** AI-powered monitoring of crop health, soil condition, and pest risks using multispectral/hyperspectral imaging and sensor data.
- **Theme-** Agriculture, Food Tech & Rural Development
- **PS Category- Software**
- **Team Name (Registered on portal)-**

Harvest Hackers.



➤ Proposed Solution:

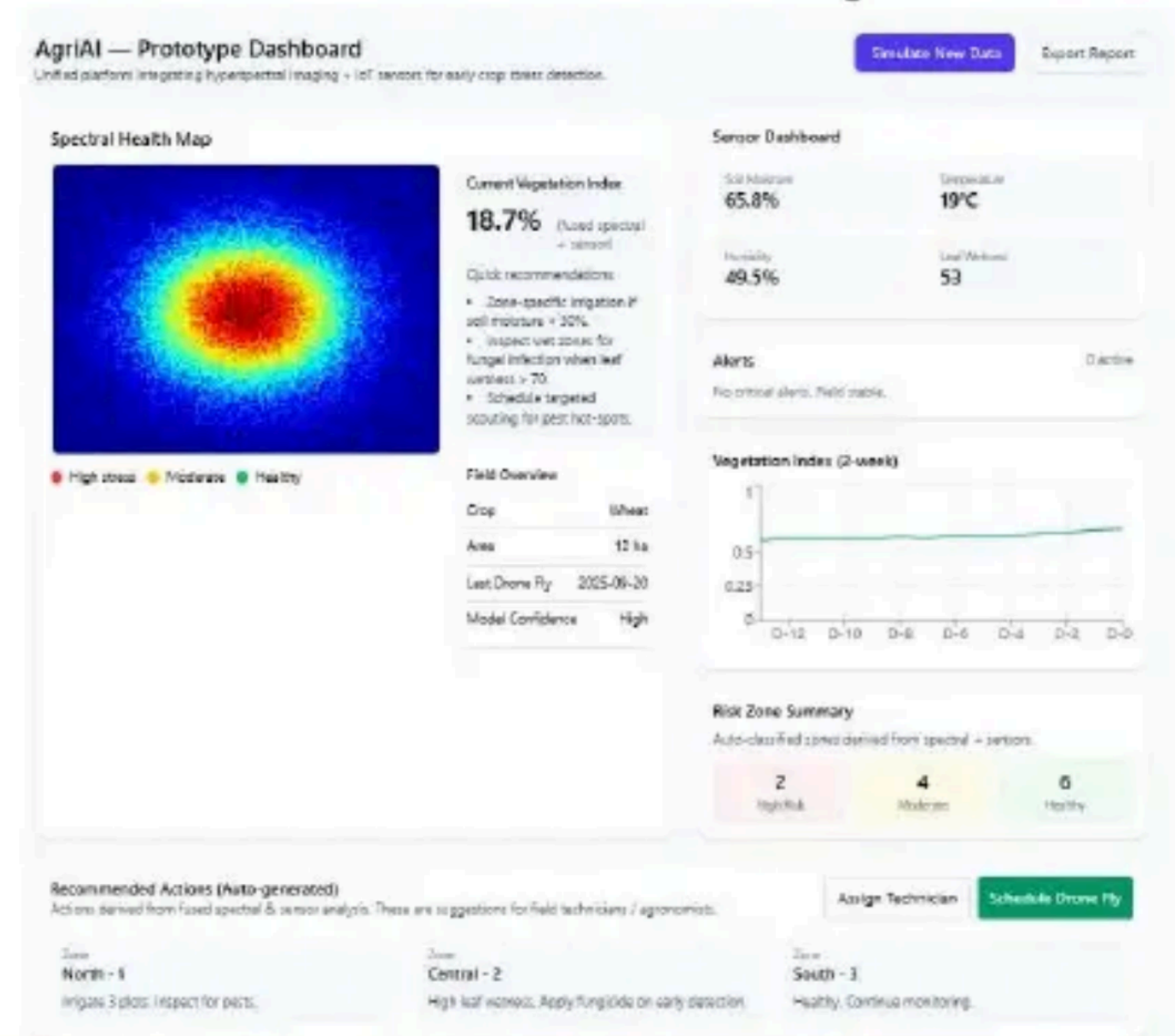
- An AI-powered software platform for modern agriculture.
- It addresses challenges like soil degradation and pest outbreaks.
- It integrates remote sensing with environmental sensor data.
- This enables a proactive farming, data driven approach to farming .

➤ How it address the problem:

- Delivers real-time insights, avoiding delays of traditional methods .
- Enables proactive management by detecting issues like plant stress early.
- Precision farming with detailed, localized data for efficient resource use.
- Prediction with AI, forecasting risks to give farmers time to prepare.

➤ Innovation & Uniqueness:

- Multispectral/hyperspectral imaging with IoT sensors for holistic crop monitoring.
- Uses AI-driven predictive analytics to detect stress, disease, and pests early.
- Provides real-time, data-based alerts instead of manual observation.
- Enables precision farming decisions to optimize resources and boost yield.



Technologies Used

Languages: Python, Java, JavaScript

Frontend: React.js (Web), Flutter (Mobile)

Backend: Node.js / Spring Boot

Databases: PostgreSQL (profiles, alerts), MongoDB (sensor & imaging data)

AI/ML: TensorFlow, PyTorch, scikit-learn, OpenCV (crop health & disease detection)

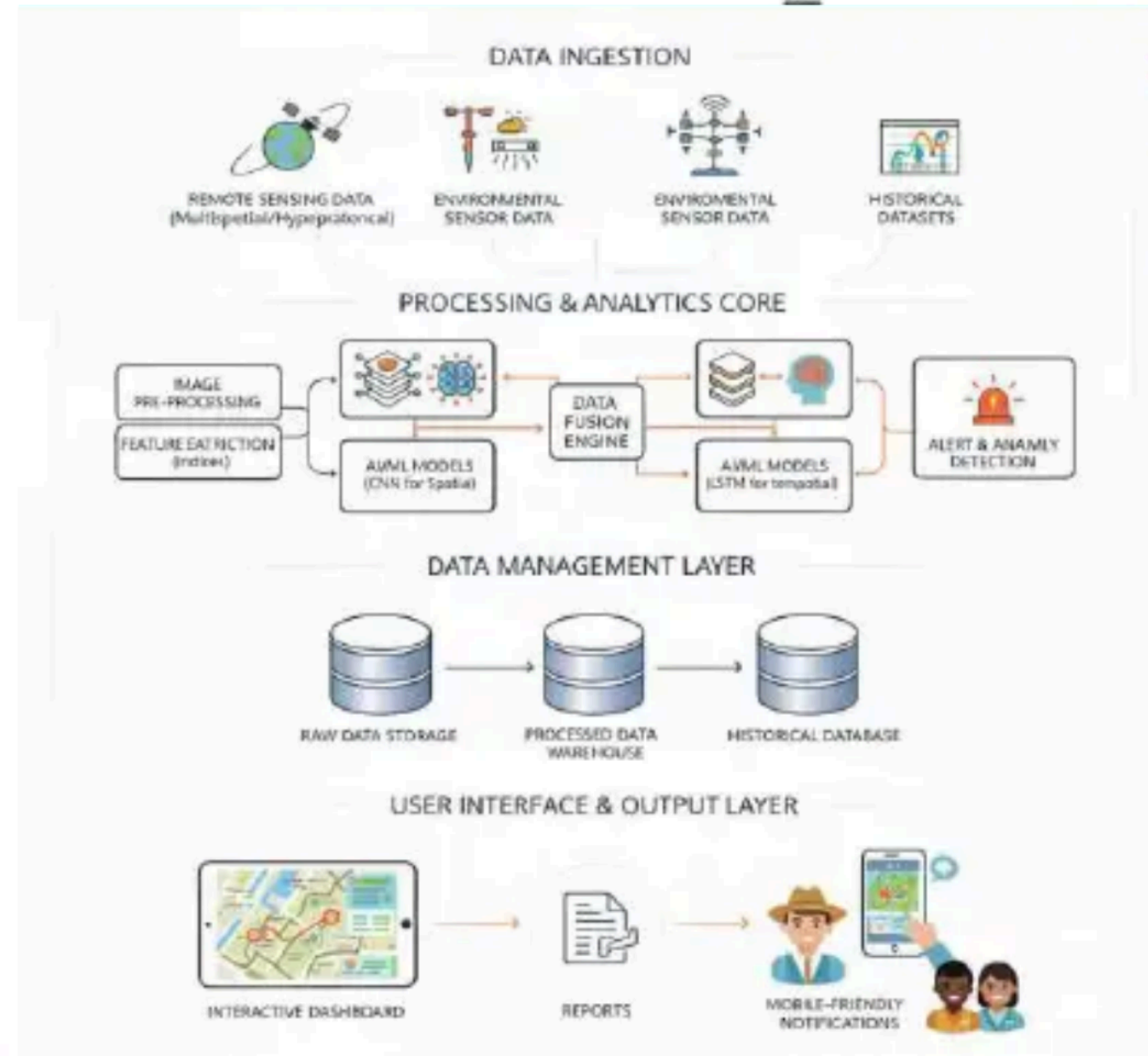
Sensors & Imaging: Multispectral/hyperspectral cameras, IoT soil sensors

Geo-Tracking: Google Maps API, GPS services

Notifications: Firebase Cloud Messaging, SMS gateway

Deployment: AWS/Azure, Docker, Kubernetes

Version Control: GitHub, GitHub Actions



➤ Feasibility Analysis:

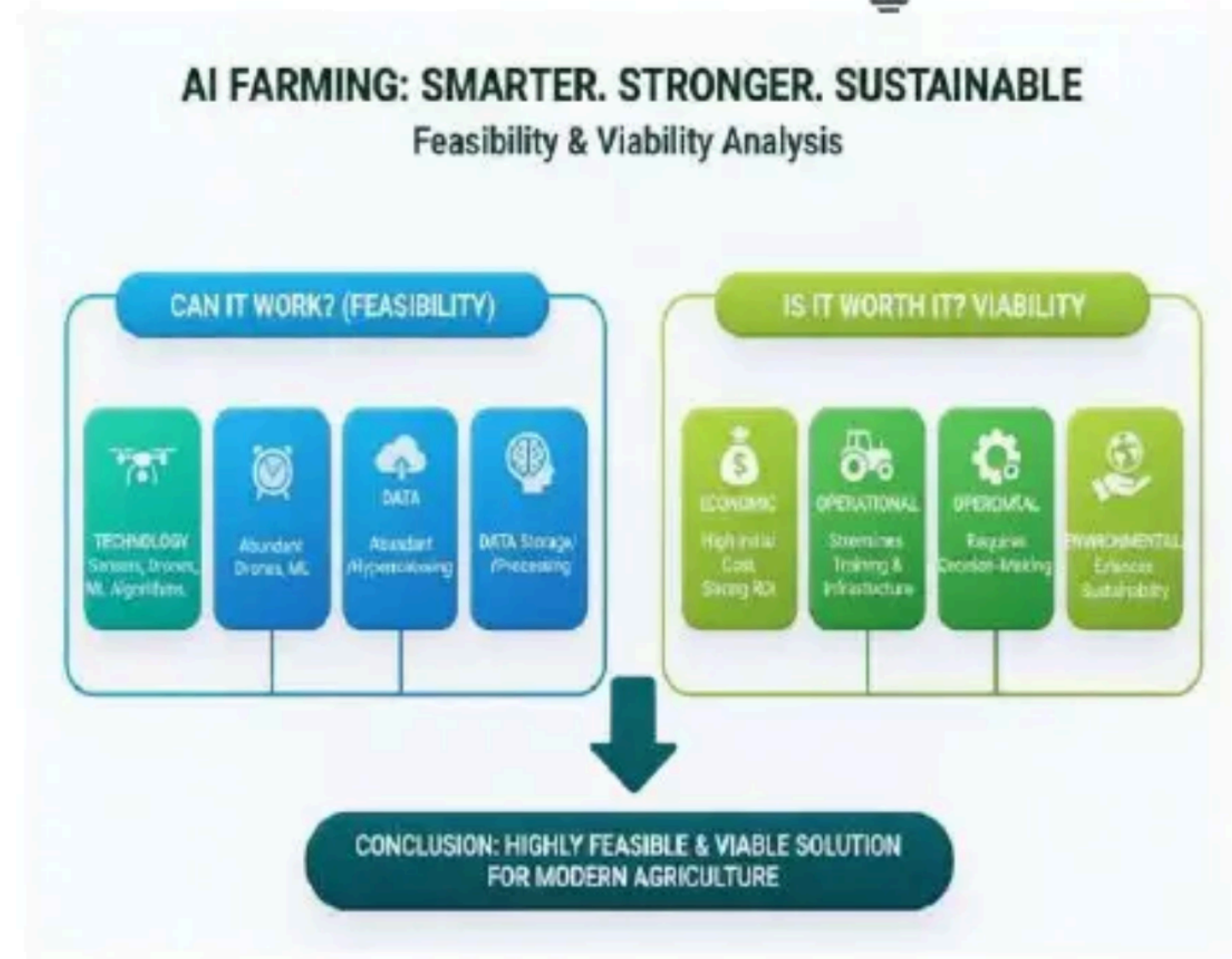
- **Technical:** Integrates AI, IoT, and cloud tech for real-time crop monitoring..
- **Economic:** Cost-effective solution with high potential ROI for farmers
- **Operational:** User-friendly apps and dashboards enable easy adoption
- **Environmental:** Reduces pesticide overuse and promotes sustainable farming.

➤ Viability:

- **Technical Viability:** AI and sensor technologies can be effectively implemented.
- **Economic Viability:** Affordable setup with long-term benefits for farmers.
- **Operational Viability:** Easy integration into existing farm practices.
- **Market Viability:** Addresses real farmer needs, ensuring adoption and impact.

➤ Business potential:

- **Market Demand:** High demand for precision agriculture solutions.
- **Revenue Potential:** Subscription-based model for farmers and agribusinesses.
- **Scalability:** Can expand to multiple regions and crop types.
- **Competitive Advantage:** Early problem detection sets it apart from traditional methods.



IMPACT AND BENEFITS



➤ Applications:

- John Deere See & Spray – Precision spraying with AI & computer vision.
- Microsoft FarmBeats – IoT + AI platform for smart farming.
- IBM Watson Decision Platform for Agriculture – Data-driven crop management.
- CropIn SmartFarm – AI-based farm management and predictive analytics.
- Taranis AI – Drone and AI-powered pest/disease detection.

➤ Books:

- Kaul, A., & Choudhary, A. (2021). *Artificial Intelligence in Agriculture: Emerging Applications*. Springer.
- Pedersen, S. M., & Lind, K. M. (2017). *Precision Agriculture: Technology and Economic Perspectives*. Springer.
- Liakos, K. G., et al. (2018). *Machine Learning in Agriculture: A Review*. Sensors Journal.

➤ Tools:

- TensorFlow / PyTorch – AI/ML frameworks used in crop prediction models.
- QGIS & ArcGIS – Geographic Information System tools for mapping farmland.
- Raspberry Pi & Arduino – IoT devices for smart farming sensors.
- Drones (DJI Agriculture Drones) – For monitoring crops & spraying.
- Remote Sensing & Satellite Data (Sentinel, Landsat) – For crop health monitoring.

